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Vehicular display control device, vehicular display device, vehicular display control method, and non-transitory recording medium

Abstract

A vehicular display control device that is configured to: display, at a display, an image of a switch that causes a transition from a first state to a second state in response to being pressed a first time and does not cause a return from the second state to the first state in response to being pressed a second time or more than a second time; and display, at the display, a text string indicating that the transition to the second state is complete in response to the switch having been pressed once in a state in which the image is displayed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2018/0178649	12/2017	Mimura	N/A	B60K 35/00
2019/0071083	12/2018	Takahashi	N/A	B60W 30/143
2020/0034102	12/2019	Hirota et al.	N/A	N/A
2020/0062276	12/2019	Yuan	N/A	B60K 35/22
2021/0138903	12/2020	Kozono	N/A	B60K 35/00
2022/0055482	12/2021	Brissman	N/A	B60K 35/10
2022/0089061	12/2021	Farley	N/A	B60W 30/182

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017-109540	12/2016	JP	N/A
2020-15453	12/2019	JP	N/A
2021-075157	12/2020	JP	N/A

OTHER PUBLICATIONS

“Drive Tech: All about the Honda Odyssey and its driver assistance features,” published by Youtube channel Carsplain on Jun. 16, 2020; video available at <https://www.youtube.com/watch?v=Y-fqr9sJhs8> (Year: 2020). cited by examiner

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2021-135742 filed on Aug. 23, 2021, the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

(2) The present disclosure relates to a vehicular display control device, a vehicular display device, a vehicular display control method, and a non-transitory recording medium recorded with a vehicular display control program.

Related Art

(3) Japanese Patent Application Laid-Open (JP-A) No. 2021-075157 discloses technology in which a layout of a switch array provided to a steering wheel of a vehicle is displayed as an image on a head-up display (HUD).

(4) A switch array provided to the vehicle steering wheel may include a switch (also referred to as a “one-way switch”) that causes a transition from a first state to a second state on being pressed a first time, and does not cause a return from the second state to the first state on being pressed a second time onward. Note that in cases in which an image of the switch array provided to the steering wheel is displayed on an HUD, a user might misunderstand the operation of the above-described one-way switch when pressed twice or more.

SUMMARY

(5) An aspect of the present disclosure is a vehicular display control device, that includes: a memory; and a processor coupled to the memory, the processor being configured to: display, at a display, an image of a switch that causes a transition from a first state to a second state in response to being pressed a first time and does not cause a return from the second state to the first state in response to being pressed a second time or more than a second time; and display, at the display, a text string indicating that the transition to the second state is complete in response to the switch having been pressed once in a state in which the image is displayed.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a block diagram illustrating a schematic configuration of a vehicular display device according to an exemplary embodiment and other relevant portions;
- (2) FIG. 2 is a schematic configuration diagram illustrating an example of an HUD;
- (3) FIG. 3 is a functional block diagram illustrating a display control ECU;
- (4) FIG. 4 is a perspective view illustrating a front section of a vehicle cabin of a vehicle as viewed from a vehicle rear side;
- (5) FIG. 5 is a diagram illustrating an enlarged section of a steering wheel;
- (6) FIG. 6 is a flowchart illustrating control processing executed by a display control ECU; and
- (7) FIG. 7 is a concept diagram illustrating an example of an image displayed on an HUD while in an ACC-ON state;
- (8) FIG. 8 is a concept diagram illustrating an example of an image displayed on the HUD when a switch has been pressed causing a transition to an ACC-OFF state from the state in FIG. 7;
- (9) FIG. 9 is a concept diagram illustrating an example of an image displayed on the HUD after a predetermined duration has elapsed from the state in FIG. 8;
- (10) FIG. 10 is a concept diagram illustrating an example of an image displayed on an HUD while in an ACC-OFF state;
- (11) FIG. 11 is a concept diagram illustrating an example of an image displayed on the HUD when a switch has been pressed causing a transition to a state of ACC activation at a latest set value from the state in FIG. 10; and
- (12) FIG. 12 is a concept diagram illustrating an example of an image displayed on the HUD after a predetermined duration has elapsed from the state in FIG. 11.

DESCRIPTION OF EMBODIMENTS

- (13) Detailed explanation follows regarding an example of an exemplary embodiment of the

present disclosure, with reference to the drawings. FIG. 1 illustrates a vehicular display device **10** according to the present exemplary embodiment. The vehicular display device **10** is installed to a vehicle, and includes an HUD **12** serving as an example of a display, and a display control electronic control unit (ECU) **22** that controls display of images by the HUD **12**. Note that the display control ECU **22** is an example of a vehicular display control device according to the present disclosure.

(14) As illustrated in FIG. 2, the HUD **12** includes a projector **14** that emits a projection beam of a display image, a mirror **16** formed of a concave mirror that reflects and projects the projection beam emitted from the projector **14** onto a windshield **18** of the vehicle, and a mirror angle adjustment actuator (ACT) **20**. Processing such as half-silvering is performed on a vehicle cabin interior surface of a range **74** (see FIG. 4) where the display image is projected onto the windshield **18**, such that a user is able to see the projected display image, and also able to see the scene outside the vehicle through the windshield **18**. The mirror angle adjustment ACT **20** enables a projection position of the display image on the windshield **18** to be changed by adjusting the angle of the mirror **16**.

(15) As illustrated in FIG. 1, the display control ECU **22** includes a central processing unit (CPU) **24** serving as an example of a hardware processor, and memory **26** such as read only memory (ROM) and random access memory (RAM). The display control ECU **22** also includes a non-volatile storage section **28** such as a hard disk drive (HDD) and/or a solid state drive (SSD), a communication interface (I/F) **30**, and an input/output I/F **31**. The CPU **24**, the memory **26**, the storage section **28**, the communication I/F **30**, and the input/output I/F **31** are connected so as to be capable of communicating with one another through an internal bus **32**.

(16) A control program **34** is stored in the storage section **28**. The control program **34** is read from the storage section **28** and expanded in the memory **26**, and the control program **34** expanded in the memory **26** is loaded and executed by the CPU **24**, such that the display control ECU **22** functions as a control section **36** illustrated in FIG. 3. Although only one CPU **24** is illustrated for simplicity, the CPU **24** may be configured of plural CPUs. Alternatively, another processor such as a GPU may be employed instead of, or in addition to, a CPU.

(17) In a state in which an image is being displayed on the HUD **12** of a switch array (such as a steering switch array **44R** or **44L**, described later) including a switch that causes a transition from a first state to a second state on being pressed a first time and does not cause a return from the second state to the first state on being pressed a second time onward, the control section **36** performs the following processing. Namely, the control section **36** displays a text string on the HUD **12** indicating completion of the transition to the second state in cases in which the switch has been pressed once. Note that the control program **34** is an example of a vehicular display control program according to the present disclosure.

(18) The display control ECU **22** is connected to a system bus **40**. Onboard equipment **42**, a right-side steering switch array **44R**, a left-side steering switch array **44L**, and a switch management ECU **80** are connected to the system bus **40**. In addition to equipment such as an air conditioner, an audio system, a car navigation system, and an audio input device, the onboard equipment **42** also includes travel assistance equipment such as an Adaptive Cruise Control (ACC) device and a Lane Tracing Assist (LTA) device. Note that ACC is an example of a cruise control of the present disclosure.

(19) As illustrated in FIG. 4, an instrument panel **72** is installed at a front section of the vehicle cabin of a vehicle **70** equipped with the vehicular display device **10** according to the present exemplary embodiment. The windshield **18** is installed at a front end portion of the instrument panel **72**, extends along a vehicle vertical direction and a vehicle width direction, and partitions between the vehicle cabin interior and vehicle cabin exterior. A steering wheel **76** is provided to a driving seat side (vehicle right side) of the instrument panel **72** through a steering column. The steering wheel **76** includes a substantially circular ring-shaped rim **76A**. A hub **76B**, configuring a

central section, is provided at an inner peripheral side of the rim **76A**. The rim **76A** and the hub **76B** are coupled together by plural (three in the present exemplary embodiment) spokes **76C**.

(20) The spokes **76C** are provided at three locations, these being between the right side of the rim **76A** and the hub **76B**, between the left side of the rim **76A** and the hub **76B**, and between the lower side of the rim **76A** and the hub **76B**. Note that the right-side steering switch array **44R** is provided to the spoke **76C** between the right side of the rim **76A** and the hub **76B**. Similarly, the left-side steering switch array **44L** is provided to the spoke **76C** between the left side of the rim **76A** and the hub **76B**.

(21) The right-side steering switch array **44R** includes a first right-side tactile switch **46**, a second right-side tactile switch **48**, and a third right-side tactile switch **50**, each of which is configured by a contact detector switch. The right-side steering switch array **44R** also includes a first right-side static sensor **52**, a second right-side static sensor **54**, and a third right-side static sensor **56**.

(22) As illustrated in FIG. 5, the first right-side tactile switch **46** has a substantially pentagonal shape with rounded corners, and includes an upper switch **46A**, a lower switch **46B**, a right-side switch **46C**, and a left-side switch **46D**. The first right-side tactile switch **46** is therefore configured so as to be capable of being pressed at upper, lower, left, and right sides.

(23) The first right-side tactile switch **46** is further provided with the first right-side static sensor **52** that is capable of detecting whether a user seated in the driving seat of the vehicle (hereafter simply referred to as the “user”) has touched the first right-side tactile switch **46**. Specifically, the first right-side static sensor **52** is capable of detecting whether the user has touched any switch out of the upper switch **46A**, the lower switch **46B**, the right-side switch **46C**, or the left-side switch **46D** of the first right-side tactile switch **46**.

(24) When the steering wheel **76** is viewed face-on, the second right-side tactile switch **48** is installed adjacent to the left side of the first right-side tactile switch **46**, is formed in a substantially rectangular shape with its length direction along the vehicle vertical direction, and includes an upper switch **48A** and a lower switch **48B**. Thus, the second right-side tactile switch **48** is configured so as to be capable of being pressed at upper and lower sides.

(25) The second right-side tactile switch **48** is further provided with the second right-side static sensor **54** that is capable of detecting whether the user has touched the second right-side tactile switch **48**. Specifically, the second right-side static sensor **54** is capable of detecting whether the user has touched any switch out of the upper switch **48A** or the lower switch **48B** of the second right-side tactile switch **48**.

(26) When the steering wheel **76** is viewed face-on, the third right-side tactile switch **50** is installed at the lower side of both the first right-side tactile switch **46** and the second right-side tactile switch **48**, and is formed in a substantially rectangular shape with its length direction along the vehicle width direction. The third right-side tactile switch **50** is provided with the third right-side static sensor **56** that is configured so as to be capable of detecting whether the user has touched the third right-side tactile switch **50**.

(27) As illustrated in FIG. 1, the left-side steering switch array **44L** includes a first left-side tactile switch **58**, a second left-side tactile switch **60**, and a third left-side tactile switch **62**, each of which is configured by a contact detector switch. The left-side steering switch array **44L** also includes a first left-side static sensor **64**, a second left-side static sensor **66**, and a third left-side static sensor **68**.

(28) As illustrated in FIG. 5, the first left-side tactile switch **58** has a substantially pentagonal shape with rounded corners, and includes an upper switch **58A**, a lower switch **58B**, a right-side switch **58C**, and a left-side switch **58D**. The first left-side tactile switch **58** is therefore configured so as to be capable of being pressed at upper, lower, left, and right sides.

(29) The first left-side tactile switch **58** is further provided with the first left-side static sensor **64** that is capable of detecting whether the user has touched the first left-side tactile switch **58**. Specifically, the first left-side static sensor **64** is capable of detecting whether the user has touched

any switch out of the upper switch **58A**, the lower switch **58B**, the right-side switch **58C**, or the left-side switch **58D** of the first left-side tactile switch **58**.

(30) When the steering wheel **76** is viewed face-on, the second left-side tactile switch **60** is installed adjacent to the right side of the first left-side tactile switch **58**, is formed in a substantially rectangular shape with its length direction along the vehicle vertical direction, and includes an upper switch **60A** and a lower switch **60B**. Thus, the second left-side tactile switch **60** is configured so as to be capable of being pressed at upper and lower sides.

(31) The second left-side tactile switch **60** is further provided with the second left-side static sensor **66** that is capable of detecting whether the user has touched the second left-side tactile switch **60**. Specifically, the second left-side static sensor **66** is capable of detecting whether the user has touched any switch out of the upper switch **60A** or the lower switch **60B** of the second left-side tactile switch **60**.

(32) When the steering wheel **76** is viewed face-on, the third left-side tactile switch **62** is installed at the lower side of both the first left-side tactile switch **58** and the second left-side tactile switch **60**, and is formed in a substantially rectangular shape with its length direction along the vehicle width direction. The third left-side tactile switch **62** is provided with the third left-side static sensor **68** that is capable of detecting whether the user has touched the third left-side tactile switch **62**.

(33) The switch management ECU **80** illustrated in FIG. **1** includes a CPU, memory, a non-volatile storage section, a communication I/F, and so on, similarly to the display control ECU **22**. A function allocation table **82** is stored in the storage section of the switch management ECU **80**. Functions allocated to the respective switches of the switch arrays **44R**, **44L** when the vehicle **70** is in various states, and conditions under which the vehicle **70** transitions to the various states, are registered in the function allocation table **82**.

(34) When any switch out of the steering switch arrays **44R**, **44L** is pressed, the switch management ECU **80** reads the function allocated to this pressed switch from the function allocation table **82**. The switch management ECU **80** then notifies the item of onboard equipment **42** corresponding to this read function that execution of the read function has been instructed by pressing the corresponding switch.

(35) For example, FIG. **7** illustrates an image of the switch array **44R** displayed on the HUD **12** while the vehicle **70** is in a first state in which the ACC is ON, and the functions allocated to the respective switches are indicated by either an icon or text. For example, a function to cause the vehicle **70** to transition from the first state of ACC-ON to a second state of ACC-OFF is allocated to the right-side switch **58C**. If for example the user presses the right-side switch **58C** while in the first state, the switch management ECU **80** notifies the ACC device of the onboard equipment **42** that a transition to the second state of ACC-OFF has been instructed.

(36) Next, explanation follows regarding control processing serving as operation of the present exemplary embodiment, executed by the display control ECU **22** while an ignition switch of the vehicle **70** is on, with reference to FIG. **6**.

(37) At step **200** of the control processing, the control section **36** determines whether or not an image of at least one switch array out of the switch arrays **44R**, **44L** is being displayed on the HUD **12**. In the present exemplary embodiment as an example, in cases in which any switch of the switch array **44R** is being touched by the user, an image of the switch array **44R** is displayed on the HUD **12**. Similarly, in cases in which any switch of the switch array **44L** is being touched by the user, an image of the switch array **44L** is displayed on the HUD **12**.

(38) Thus, in cases in which none of the switches out of the switch array **44R** and the switch array **44L** are being touched, a negative determination is made at step **200**, processing returns to step **200**, and the determination of step **200** is repeated. In such cases, no image of the switch arrays **44R**, **44L** is displayed on the HUD **12**. On the other hand, in cases in which at least one switch out of the switch arrays **44R**, **44L** is being touched, an affirmative determination is made at step **200** and processing proceeds to step **202**. In such cases, an image of whichever switch array **44** is being

touched out of the switch arrays **44R**, **44L** is set as a display target.

(39) Note that after this image of the touched switch array **44** has been temporarily displayed on the HUD **12**, display of this image may be removed as soon as this touch-contact is released, or display of this image may be removed at a timing when a predetermined duration has elapsed since the touch-contact was released.

(40) At step **202**, the control section **36** acquires information (information registered in the function allocation table **82**) from the switch management ECU **80** indicating a function allocation status of the respective switches of the display target switch array (at least one switch array out of the switch arrays **44R**, **44L**).

(41) At step **204**, the control section **36** generates an image of the display target switch array (the at least one switch array out of the switch arrays **44R**, **44L**) based on the information acquired at step **202**. Namely, first, the control section **36** reads data regarding an image representing the display target switch array **44** from the storage section **28** or the like. Next, for any switch that has been allocated a press-implemented function amongst the respective switches of the display target switch array **44**, the control section **36** depicts an icon or text corresponding to this allocated function in a display region corresponding to the switch in the read image.

(42) As an example, FIG. **7** illustrates a state in which icons or text corresponding to the allocated functions are depicted in display regions corresponding to the respective switches of the switch array **44R** that is the display target switch array while in the ACC-ON state. As another example, FIG. **10** illustrates a state in which icons or text corresponding to the allocated functions are depicted in display regions corresponding to the respective switches of the switch array **44R** that is the display target switch array while in the ACC-OFF state.

(43) Next, the control section **36** combines an image generated by the above-described processing (an image representing the display target switch array **44**) with other images displayed on the HUD **12** (such as images indicating the vehicle speed and shift position as illustrated in FIG. **7**). The control section **36** then displays the combined images on the HUD **12**.

(44) Next, at step **206**, the control section **36** determines whether or not any switch out of the display target switch array **44** has been touched by the user. In cases in which a negative determination is made at step **206**, the processing of step **206** is repeated until an affirmative determination is made. In cases in which an affirmative determination is made at step **206**, processing proceeds to step **208**. At step **208**, the control section **36** inverts the display colors of the icon or text displayed in the display region corresponding to the touched switch.

(45) As an example, FIG. **7** illustrates a case in which the right-side switch **46C** of the switch array **44R** that is the display target switch array has been touched while in the ACC-ON state, and so the display colors of a text string "CANCEL" displayed in the display region corresponding to the right-side switch **46C** are inverted. As another example, FIG. **10** illustrates a case in which the upper switch **46A** of the switch array **44R** that is the display target switch array has been touched while in the ACC-OFF state, and so the display colors of a text string "RES" displayed in the display region corresponding to the upper switch **46A** are inverted. Inverting the display colors in the display region corresponding to the switch touched by user in this manner enables the user to be made aware of the touch position touched by the user.

(46) At step **210**, the control section **36** determines whether or not the switch that was touched by the user has also been pressed by the user. For example, in cases in which the user has moved their fingers away from the touched switch without pressing this switch, a negative determination is made at step **210** and processing returns to step **206**. On the other hand, in cases in which the user has also pressed the touched switch, an affirmative determination is made at step **210** and processing proceeds to step **212**.

(47) At step **212**, the control section **36** determines whether or not the switch pressed by the user is a switch for causing a transition from the ACC-ON state to the ACC-OFF state. In the present exemplary embodiment, the switch for causing a transition from the ACC-ON state to the ACC-

OFF state while in the ACC-ON state illustrated in FIG. 7 is the right-side switch **46C** for which a text string “CANCEL” is displayed in the corresponding display region. In this example, the right-side switch **46C** is a one-way switch that enables a transition from the first state of ACC-ON to the second state of ACC-OFF on being pressed a first time, but does not cause a return from the second state of ACC-OFF to the first state of ACC-ON on being pressed a second time onward. More specifically, in this example, the right-side switch **46C** is a one-way switch that has not been allocated any function for being pressed a second time onward.

(48) Thus, in cases in which right-side switch **46C** for which the text string “CANCEL” is displayed in the corresponding display region has been pressed while in the ACC-ON state illustrated in FIG. 7, an affirmative determination is made at step **212** and processing proceeds to step **214**. At step **214**, the control section **36** displays a text string “CANCELED” (see the text string **90** illustrated in FIG. 8), indicating that the right-side switch **46C** has been pressed once and the transition to the second state of ACC-OFF is complete, in a display region at the upper side of a region where the vehicle speed and so on are displayed on the HUD **12**. This enables the user to be made aware based on the text string “CANCELED” displayed on the HUD **12** that a return from the second state of ACC-OFF to the first state of ACC-ON cannot be realized by further pressing of the right-side switch **46C**.

(49) At step **216**, the control section **36** displays a circular first symbol **92** (see FIG. 8), indicating that the right-side switch **46C** is a switch that has not been allocated any function for being pressed a second time onward, in the display region corresponding to the right-side switch **46C** on the HUD **12**. This enables the user to be made aware of the operation of the right-side switch **46C** when pressed twice or more, namely that the right-side switch **46C** causes a transition from the first state of ACC-ON to the second state of ACC-OFF on being pressed a first time, and does not cause a return from the second state of ACC-OFF to the first state of ACC-ON on being pressed a second time onward.

(50) After the processing of step **216**, processing proceeds to step **226**. At step **226**, the control section **36** determines whether or not a predetermined duration has elapsed since the text string was initially displayed on the HUD **12** at step **214**. Note that the predetermined duration in the determination regarding step **214** is for example a duration of approximately several seconds (such as two seconds). In cases in which a negative determination is made at step **226**, the processing of step **226** is repeated until an affirmative determination is made.

(51) In cases in which an affirmative determination is made at step **226**, processing proceeds to step **228**. At step **228**, the control section **36** removes the text string displayed on the HUD **12** at step **214** from the HUD **12** (see FIG. 9 also). After the processing of step **228**, processing returns to step **200**.

(52) On the other hand, in cases in which the switch pressed by the user is not a switch for causing a transition from the ACC-ON state to the ACC-OFF state, a negative determination is made at step **212** and processing proceeds to step **218**. At step **218**, the control section **36** determines whether or not the switch pressed by the user is a switch for causing a transition from the ACC-OFF state to a state in which the ACC is activated at a latest set value (such as at a last set vehicle speed).

(53) In the present exemplary embodiment, the switch for causing a transition from the ACC-OFF state to the state of ACC activation at the latest set value while in the ACC-OFF state as illustrated in FIG. 10 is the upper switch **46A** for which a text string “RES” is displayed in the corresponding display region. In this example, the upper switch **46A** is a one-way switch that enables a transition from a first state of ACC-OFF to a second state of ACC activation at the latest set value on being pressed a first time, but does not cause a return from the second state of ACC activation at the latest set value to the first state of ACC-OFF on being pressed a second time onward. More specifically, in this example, the upper switch **46A** is a one-way switch that has been allocated a function other than a function to cause a transition from the second state of ACC activation at the latest set value to the first state of ACC-OFF for being pressed a second time onward, namely a function to

increase a set vehicle speed during ACC.

(54) Thus, in cases in which the upper switch **46A** for which the text string “RES” is displayed in the corresponding display region has been pressed while in the ACC-OFF state illustrated in FIG. **10**, an affirmative determination is made at step **218** and processing proceeds to step **220**. At step **220**, the control section **36** displays a text string “RESUMED” (see the text string **94** in FIG. **11**), indicating that the upper switch **46A** has been pressed once and the transition to the second state of ACC activation at the latest set value is complete, in the display region at the upper side of the region where the vehicle speed and so on are displayed on the HUD **12**. This enables the user to be made aware based on the text string “RESUMED” displayed on the HUD **12** that a return from the second state of ACC activation at the latest set value to the first state of ACC-OFF cannot be realized by further pressing of the upper switch **46A**.

(55) Furthermore, at step **222**, the control section **36** displays a second symbol **96** (see FIG. **11**), indicating that the upper switch **46A** is a switch that has been allocated a function to increase the set vehicle speed during ACC when pressed a second time onward, in the display region corresponding to the upper switch **46A** on the HUD **12**. This enables the user to be made aware of operation of the upper switch **46A** when pressed twice or more, namely that the upper switch **46A** causes a transition from the first state of ACC-OFF to the second state of ACC activation at the latest set value on being pressed a first time, and does not cause a return from the second state of ACC activation at the latest set value to the first state of ACC-OFF on being pressed a second time onward.

(56) After the processing of step **222**, processing proceeds to step **226**. At step **226**, the control section **36** determines whether or not a predetermined duration has elapsed since the text string was initially displayed on the HUD **12** at step **220**. In cases in which a negative determination is made at step **226**, the processing of step **226** is repeated until an affirmative determination is made. In cases in which an affirmative determination is made at step **226**, processing proceeds to step **228**. At step **228**, the control section **36** removes the text string displayed on the HUD **12** at step **220** from the HUD **12** (see FIG. **12** also). After the processing of step **228**, processing returns to step **200**.

(57) Note that in cases in which the switch pressed by the user is not a switch for causing a transition from the ACC-OFF state to the state of ACC activation at the latest set value, a negative determination is made at step **218** and processing proceeds to step **224**. At step **224**, the control section **36** performs processing corresponding to this other pressed switch.

(58) Thus, in the present exemplary embodiment, in a state in which an image is being displayed on the HUD **12** of a switch that causes a transition from a first state to a second state on being pressed a first time, and that does not cause a return from the second state to the first state on being pressed a second time onward, the control section **36** performs the following processing. Namely, in cases in which the switch that does not cause a return from the second state to the first state on being pressed a second time onward has been pressed once, the control section **36** displays a text string on the HUD **12** indicating that the transition to the second state is complete. This enables the user to be made aware based on the text string displayed on the HUD **12** that a return from the first state to the second state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state to the second state.

(59) Moreover, in the present exemplary embodiment, a switch that does not cause a return from the second state to the first state on being pressed a second time onward is a switch for causing a transition from a first state of ACC-ON to a second state of ACC-OFF. This enables the user to be made aware that a return from the ACC-OFF state to the ACC-ON state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state of ACC-ON to the second state of ACC-OFF.

(60) Moreover, in the present exemplary embodiment, the text string indicating that the transition to the second state is complete is “CANCELED”. This enables the user to intuitively ascertain that

a return from the ACC-OFF state to the ACC-ON state cannot be realized by further pressing of the pressed switch.

(61) Moreover, in the present exemplary embodiment, another switch that does not cause a return from the second state to the first state on being pressed a second time onward is a switch for causing a transition from a first state of ACC-OFF to a second state of ACC activation at the latest set value. This enables the user to be made aware that a return from the state of ACC activation at the latest set value to the ACC-OFF state cannot be realized by further pressing of the switch that was pressed to cause the transition from the ACC-OFF state to the state of ACC activation at the latest set value.

(62) Furthermore, in the present exemplary embodiment, the text string indicating that the transition to the second state is complete is “RESUMED”. This enables the user to intuitively ascertain that a return from the state of ACC activation at the latest set value to the ACC-OFF state cannot be realized by further pressing of the pressed switch.

(63) Moreover, in the present exemplary embodiment, the control section **36** removes the text string indicating that the transition to the second state is complete after a predetermined duration has elapsed since the text string was initially displayed on the HUD **12**. This enables any annoyance toward the display on the HUD **12** to be lessened compared to cases in which the text string continues to be displayed on the HUD **12** after this predetermined duration.

(64) Moreover, in the present exemplary embodiment, the HUD is employed as the display. Thus, the amount by which the user has to shift their gaze from a state in which the user is looking ahead of the vehicle through the windshield in order to view the image displayed on the display can be reduced compared to cases in which the display is a meter display or the like.

(65) Note that a case in which the ACC-ON state is the first state and the ACC-OFF state is the second state, and a case in which the ACC-OFF state is the first state and the state of ACC activation at the latest set value is the second state, have been described above. However, the first state and the second state of the present disclosure are not limited to these respective states, and may be other states.

(66) Moreover, although a format has been described in which “CANCELED” and “RESUMED” are employed as examples of “a text string indicating that the transition to the second state is complete in cases in which the switch has been pressed once”, there is no limitation thereto, and other text strings may be employed.

(67) Moreover, although a case has been described in which the HUD **12** is employed as an example of the display of the present disclosure, the present disclosure is not limited thereto. The display of the present disclosure may be another display such as a meter display.

(68) Moreover, a case has been described in which the control program **34** that is an example of the vehicular display control program according to the present disclosure is pre-stored (installed) in the storage section **28**. However, the vehicular display control program according to the present disclosure may be provided in a format recorded in a non-transitory recording medium such as an HDD, an SSD, or a DVD.

(69) The present disclosure provides a vehicular display control device, a vehicular display device, a vehicular display control method, and a non-transitory computer-readable recording medium recorded with a vehicular display control program that enable a user to be made aware that a return from a second state to a first state cannot be realized by further pressing of a switch that was pressed to cause a transition from the first state to the second state.

(70) A first aspect of the present disclosure is a vehicular display control device, that includes: a memory; and a processor coupled to the memory, the processor being configured to: display, at a display, an image of a switch that causes a transition from a first state to a second state in response to being pressed a first time and does not cause a return from the second state to the first state in response to being pressed a second time or more than a second time; and display, at the display, a text string indicating that the transition to the second state is complete in response to the switch

having been pressed once in a state in which the image is displayed.

(71) In the first aspect, the image of the switch that causes a transition from the first state to the second state on being pressed a first time and does not cause a return from the second state to the first state on being pressed a second time onward is displayed on the display. Note that the text string indicating that the transition to the second state is complete is displayed on the display in cases in which the switch has been pressed once. This enables a user to be made aware based on the text string displayed on the display that a return from the first state to the second state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state to the second state.

(72) A second aspect of the present disclosure is the vehicular display control device of the first aspect, wherein the switch is configured to cause a transition from the first state, in which cruise control is ON, to the second state, in which the cruise control is OFF.

(73) The second aspect enables the user to be made aware that a return from the cruise control-OFF state to the cruise control-ON state cannot be realized by further pressing of the switch that was pressed to cause the transition from the cruise control-ON state to the cruise control-OFF state.

(74) A third aspect of the present disclosure is the vehicular display control device of the second aspect, wherein the text string represents “canceled”.

(75) The third aspect enables the user to intuitively ascertain that a return from the cruise control-OFF state to the cruise control-ON state cannot be realized by further pressing of the pressed switch.

(76) A fourth aspect of the present disclosure is the vehicular display control device of the first aspect, wherein the switch is for causing a transition from the first state, in which cruise control is OFF, to the second state, in which the cruise control is activated at a most recent setting value.

(77) The fourth aspect enables the user to be made aware that a return from the state of cruise control activation at the latest set value to the cruise control-OFF state cannot be realized by further pressing of the switch that was pressed to cause the transition from the cruise control-OFF state to the state of cruise control activation at the latest set value.

(78) A fifth aspect of the present disclosure is the vehicular display control device of the fourth aspect, wherein the text string represents “resumed”.

(79) The fifth aspect enables the user to intuitively ascertain that a return from the state of cruise control activation at the latest set value to the cruise control-OFF state cannot be realized by further pressing of the pressed switch.

(80) A sixth aspect of the present disclosure is the vehicular display control device of any of the first to fifth aspect, wherein the processor is configured to erase display of the text string after a predetermined duration has elapsed since the text string was initially displayed on the display.

(81) The sixth aspect enables any annoyance toward the display on the display to be lessened compared to cases in which the text string continues to be displayed on the display after this predetermined duration.

(82) A seventh aspect of the present disclosure is the vehicular display control device of any of the first to sixth aspect, wherein the display is a head-up display.

(83) In the seventh aspect, since the display is the HUD, the amount by which the user has to shift their gaze from a state in which the user is looking ahead of the vehicle through a windshield in order to view the image displayed on the display can be reduced compared to cases in which the display is a meter display or the like.

(84) An eighth aspect of the present disclosure is a vehicular display device, that includes: the vehicular display control device of any of the first to seventh aspect; and the display.

(85) Similarly to the first aspect, the eighth aspect enables the user to be made aware that a return from the first state to the second state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state to the second state.

(86) A ninth aspect of the present disclosure is a vehicular display control method. The method

includes, by a processor: displaying, at a display, an image of a switch that causes a transition from a first state to a second state in response to being pressed a first time and does not cause a return from the second state to the first state in response to being pressed a second time or more than a second time; and displaying, at the display, a text string indicating that the transition to the second state is complete in response to the switch having been pressed once in a state in which the image is displayed.

(87) Similarly to the first aspect, the ninth aspect enables the user to be made aware that a return from the first state to the second state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state to the second state.

(88) A tenth aspect of the present disclosure is a non-transitory computer-readable storage medium storing a vehicular display control program executable by a computer to perform vehicular display control processing. The processing includes: displaying, at a display, an image of a switch that causes a transition from a first state to a second state in response to being pressed a first time and does not cause a return from the second state to the first state in response to being pressed a second time or more than a second time; and displaying, at the display, a text string indicating that the transition to the second state is complete in response to the switch having been pressed once in a state in which the image is displayed.

(89) Similarly to the first aspect, the tenth aspect enables the user to be made aware that a return from the first state to the second state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state to the second state.

(90) The present disclosure exhibits an advantageous effect of enabling the user to be made aware that a return from the first state to the second state cannot be realized by further pressing of the switch that was pressed to cause the transition from the first state to the second state.

Claims

1. A vehicular display device comprising: a switch array including a plurality of switches located on a steering wheel of a vehicle; a display provided in the vehicle at a location that is separate from the steering wheel, the display displaying a graphical representation of the switch array that includes information at areas of the graphical representation respectively corresponding to each of the plurality of switches, the information identifying a function that is to be performed when a user presses the switch at a location of the switch array that corresponds to a location of the area of the graphical representation having the information; a memory; and a processor coupled to the memory, the processor controlling the information to be displayed on the display, and the processor being configured to: in a state in which a status of a cruise control of the vehicle is an ON state, (1) display information representing an OFF state of the cruise control of the vehicle, the information representing the OFF state of the cruise control being displayed at a first area amongst the areas of the graphical representation of the switch array, a location of the first area within the graphical representation corresponding to a location of a first switch of the plurality of switches within the switch array that causes the cruise control to transition from the ON state to the OFF state, and (2) while the information representing the OFF state of the cruise control of the vehicle is being displayed at the first area, (2a) display at a second area amongst the areas of the graphical representation of the switch array, text or an icon representing a cruise control function that is performed when a second switch of the plurality of switches of the switch array is pressed, a location of the second area within the graphical representation corresponding to a location of the second switch within the switch array, and (2b) display at a third area amongst the areas of the graphical representation of the switch array, text or an icon representing a different cruise control function that is performed when a third switch of the plurality of switches of the switch array is pressed, a location of the third area within the graphical representation corresponding to a location of the third switch within the switch array; and in response to the first switch having been pressed:

(i) display, at a predefined area of the display, a text string indicating that a transition of the cruise control from the ON state to the OFF state is complete, (ii) erase the display of the information representing the OFF state at the first area of the graphical representation of the switch array, (iii) change the text or the icon that is displayed at the second area of the graphical representation of the switch array to indicate that the cruise control function that is performed when the second switch of the switch array is pressed has changed due to the transition of the cruise control from the ON state to the OFF state, and (iv) change the text or the icon that is displayed at the third area of the graphical representation of the switch array to indicate that the different cruise control function that is performed when the third switch of the switch array is pressed has changed due to the transition of the cruise control from the ON state to the OFF state.

2. The vehicular display device of claim 1, wherein the text string represents “canceled”.

3. The vehicular display device of claim 1, wherein when the cruise control has transitioned to the OFF state, pressing of the second switch causes the cruise control to transition from the OFF state to the ON state, in which the cruise control is activated at a most recent setting value and, when the cruise control is in the OFF state, the text or the icon that is displayed at the second area of the graphical representation of the switch array indicates that pressing of the second switch will cause the cruise control to transition to the ON state.

4. The vehicular display device of claim 3, wherein in response to the second switch being pressed to transition the cruise control from the OFF state to the ON state, the processor causes the text string that is displayed at the predefined area to represent “resumed”.

5. The vehicular display device of claim 1, wherein the processor is configured to erase display of the text string after a predetermined duration has elapsed since the text string was initially displayed at the predefined area of the display.

6. The vehicular display device of claim 1, wherein the display is a head-up display.

7. The vehicular display device of claim 1, wherein the first switch does not perform any function when the cruise control has transitioned to the OFF state, and when the cruise control has transitioned to the OFF state, a symbol indicating that the first switch does not perform any function is displayed at the first area.

8. The vehicular display device of claim 7, wherein the symbol is a circular symbol with no text.

9. A vehicular display control method in a vehicle having (i) a switch array including a plurality of switches located on a steering wheel of the vehicle, and (ii) a display provided in the vehicle at a location that is separate from the steering wheel, the display displaying a graphical representation of the switch array that includes information at areas of the graphical representation respectively corresponding to each of the plurality of switches, the information identifying a function that is to be performed when a user presses the switch at a location of the switch array that corresponds to a location of the area of the graphical representation having the information, the method comprising, by a processor: in a state in which a status of a cruise control of the vehicle is an ON state, (1) displaying information representing an OFF state of the cruise control of the vehicle, the information representing the OFF state of the cruise control being displayed at a first area amongst the areas of the graphical representation of the switch array, a location of the first area within the graphical representation corresponding to a location of a first switch of the plurality of switches within the switch array that causes the cruise control to transition from the ON state to the OFF state, and (2) while the information representing the OFF state of the cruise control of the vehicle is being displayed at the first area, (2a) displaying at a second area amongst the areas of the graphical representation of the switch array, text or an icon representing a cruise control function that is performed when a second switch of the plurality of switches of the switch array is pressed, a location of the second area within the graphical representation corresponding to a location of the second switch within the switch array, and (2b) displaying at a third area amongst the areas of the graphical representation of the switch array, text or an icon representing a different cruise control function that is performed when a third switch of the plurality of switches of the switch array is

pressed, a location of the third area within the graphical representation corresponding to a location of the third switch within the switch array; and in response to the first switch having been pressed: (i) displaying, at a predefined area of the display, a text string indicating that a transition of the cruise control from the ON state to the OFF state is complete, (ii) erasing the display of the information representing the OFF state at the first area of the graphical representation of the switch array, (iii) changing the text or the icon that is displayed at the second area of the graphical representation of the switch array to indicate that the cruise control function that is performed when the second switch of the switch array is pressed has changed due to the transition of the cruise control from the ON state to the OFF state, and (iv) changing the text or the icon that is displayed at the third area of the graphical representation of the switch array to indicate that the different cruise control function that is performed when the third switch of the switch array is pressed has changed due to the transition of the cruise control from the ON state to the OFF state.

10. The vehicular display control method of claim 9, wherein when the cruise control has transitioned to the OFF state, pressing of the second switch causes the cruise control to transition from the OFF state to the ON state, in which the cruise control is activated at a most recent setting value and, when the cruise control is in the OFF state, the text or the icon that is displayed at the second area of the graphical representation of the switch array indicates that pressing of the second switch will cause the cruise control to transition to the ON state.

11. The vehicular display control method of claim 9, wherein the first switch does not perform any function when the cruise control has transitioned to the OFF state, and when the cruise control has transitioned to the OFF state, displaying, at the first area, a symbol indicating that the first switch does not perform any function.

12. The vehicular display control method of claim 11, wherein the symbol is a circular symbol with no text.

13. A non-transitory computer-readable storage medium storing a vehicular display control program executable by a computer to perform vehicular display control processing in a vehicle having (i) a switch array including a plurality of switches located on a steering wheel of the vehicle, and (ii) a display provided in the vehicle at a location that is separate from the steering wheel, the display displaying a graphical representation of the switch array that includes information at areas of the graphical representation respectively corresponding to each of the plurality of switches, the information identifying a function that is to be performed when a user presses the switch at a location of the switch array that corresponds to a location of the area of the graphical representation having the information, the processing comprising: in a state in which a status of a cruise control of the vehicle is an ON state, (1) displaying information representing an OFF state of the cruise control of the vehicle, the information representing the OFF state of the cruise control being displayed at a first area amongst the areas of the graphical representation of the switch array, a location of the first area within the graphical representation corresponding to a location of a first switch of the plurality of switches within the switch array that causes the cruise control to transition from the ON state to the OFF state, and (2) while the information representing the OFF state of the cruise control of the vehicle is being displayed at the first area, (2a) displaying at a second area amongst the areas of the graphical representation of the switch array, text or an icon representing a cruise control function that is performed when a second switch of the plurality of switches of the switch array is pressed, a location of the second area within the graphical representation corresponding to a location of the second switch within the switch array, and (2b) displaying at a third area amongst the areas of the graphical representation of the switch array, text or an icon representing a different cruise control function that is performed when a third switch of the plurality of switches of the switch array is pressed, a location of the third area within the graphical representation corresponding to a location of the third switch within the switch array; and in response to the first switch having been pressed: (i) displaying, at a predefined area of the display, a text string indicating that a transition of the cruise control from the ON state to the OFF

state is complete, (ii) erasing the display of the information representing the OFF state at the first area of the graphical representation of the switch array, (iii) changing the text or the icon that is displayed at the second area of the graphical representation of the switch array to indicate that the cruise control function that is performed when the second switch of the switch array is pressed has changed due to the transition of the cruise control from the ON state to the OFF state, and (iv) changing the text or the icon that is displayed at the third area of the graphical representation of the switch array to indicate that the different cruise control function that is performed when the third switch of the switch array is pressed has changed due to the transition of the cruise control from the ON state to the OFF state.

14. The non-transitory computer-readable storage medium of claim 13, wherein when the cruise control has transitioned to the OFF state, pressing of the second switch causes the cruise control to transition from the OFF state to the ON state, in which the cruise control is activated at a most recent setting value and, when the cruise control is in the OFF state, the text or the icon that is displayed at the second area of the graphical representation of the switch array indicates that pressing of the second switch will cause the cruise control to transition to the ON state.

15. The non-transitory computer-readable storage medium of claim 13, wherein the first switch does not perform any function when the cruise control has transitioned to the OFF state, and when the cruise control has transitioned to the OFF state, displaying, at the first area, a symbol indicating that the first switch does not perform any function.

16. The non-transitory computer-readable storage medium of claim 15, wherein the symbol is a circular symbol with no text.
